**GANITA PRAKASH**

CLASS 7 MATHEMATICS | PART I

# A PEEK BEYOND THE POINT

Solved textbook exercises + illustrated notes

**A number line helps us see decimals**




**Questions first → worked explanations → visual notes → revision**

Aligned with the supplied NCERT *Ganita Prakash*, Grade 7, Part I.
Original explanations and original TikZ illustrations; editable in Overleaf.

# Contents

**1 Textbook exercises: worked answers first** **2**

**2 Textbook companion: complete answer reference** **4**

**3 Illustrated notes: decimal place value** **25**
3.1 What a decimal really means . . . 25
3.2 An editable number line . . . 25
3.3 Compare decimals by aligning places . . . 25
3.4 Addition and subtraction . . . 25
3.5 Measurement and money . . . 25

**4 Last-minute revision** **25**

**5 Mini-lab: grids for tenths and hundredths** **26**

**6 Why trailing zeroes are useful** **26**

**7 Quick self-check (with answers)** **26**

# Textbook exercises: worked answers first

**Decimals are fractions written by place value**

The following worked solutions cover the chapter's main exercise types. Additional page-by-page solutions from the supplied chapter appendix appear immediately afterward; the original diagram is needed where a question asks you to read a ruler or number line.

**Textbook question + answer**

**Textbook p. 47–49.** Reading tenths: $2\frac{7}{10}$ cm, $3\frac{2}{10}$ cm, and 34 tenths. **Answer:** $2\frac{7}{10} = 2.7$ cm; $3\frac{2}{10} = 3.2$ cm; $34 \times \frac{1}{10} = 3.4$. A ruler divided into ten equal parts gives tenths of a centimetre.

**Textbook question + answer**

**Textbook p. 50–53.** Add and subtract mixed tenths and hundredths. **Answer:** $2\frac{3}{10} + 5\frac{4}{10} + 7\frac{5}{10} = 15\frac{2}{10} = 15.2$. $2\frac{4}{10} - \frac{9}{10} = 1\frac{5}{10} = 1.5$. $\frac{1}{10} = \frac{10}{100}$, so $4\frac{45}{100} = 4.45$.

**Textbook question + answer**

**Textbook p. 58.** Hundredths and tenths: find $\frac{3}{10} + 3\frac{4}{100}$. **Answer:** $\frac{30}{100} + \frac{304}{100} = \frac{334}{100} = 3.34$. Make the *unit of fractional parts* the same before adding.

**Textbook question + answer**

**Textbook p. 64–69.** Metric conversions and money. **Answer:** 5.6 cm = 56 mm, 15 cm = 0.15 m, 465 g = 0.465 kg and 75 paise = Rs. 0.75. Converting to smaller units multiplies; converting to larger units divides.

**Textbook question + answer**

**Textbook p. 70–73.** Place-value comparison: Which is greater, 1.23 or 1.32? 3.81 or 13.800? **Answer:** $1.32 > 1.23$ because 3 tenths exceed 2 tenths; $13.800 > 3.81$ because $13 > 3$ in the whole-number part. Appending trailing zeroes does not change a decimal: $4.5 = 4.50$.

**Textbook question + answer**

**Textbook p. 75–77.** Calculate by aligning decimal points. **Answer:** $2.7+3.5 = 6.2$; $10.4-4.5 = 5.9$; $34.505 - 18.1 = 34.505 - 18.100 = 16.405$; $9.9 - 9.09 = 9.90 - 9.09 = 0.81$.

**Textbook question + answer**

**Textbook p. 78.** Convert $\frac{5}{100}, \frac{16}{1000}, \frac{12}{10}, \frac{254}{1000}$ into decimals. **Answer:** 0.05, 0.016, 1.2, 0.254 respectively.

**Textbook question + answer**

**Textbook p. 79.** Mahi's grocery weights and Pinto's milk deliveries. **Answer:** $0.25 + 0.30 + 0.50 + 0.20 + 0.05 = 1.30$ kg. Pinto supplied $3.79 + 4.20 + 4.25 = 12.24$ L in the first three days; remaining days $= 25 - 12.24 = 12.76$ L.

**Textbook question + answer**

**Textbook p. 79.** Weight change and large conversions. **Answer:** $35.75 - 34.50 = 1.25$ kg **lost**. 1 km = 1000 m = $10^6$ mm. One lakh passengers paying 45 paise: $100{,}000 \times 0.45 = $ Rs. 45,000.

### Textbook question + answer

**Textbook p. 80.** Write basic fractions in decimal form. **Answer:** $\frac{1}{2} = 0.5$, $\frac{3}{2} = 1.5$, $\frac{1}{4} = 0.25$, $\frac{3}{4} = 0.75$, $\frac{1}{5} = 0.2$, $\frac{4}{5} = 0.8$.

### Common Mistake

Do not compare decimal numbers only by how many digits they have: $0.9 > 0.1234$. Compare ones, tenths, hundredths, *then* thousandths. In cricket, "5.5 overs" means five overs and five balls, **not** 5.5 decimal overs.

# Textbook companion: complete answer reference

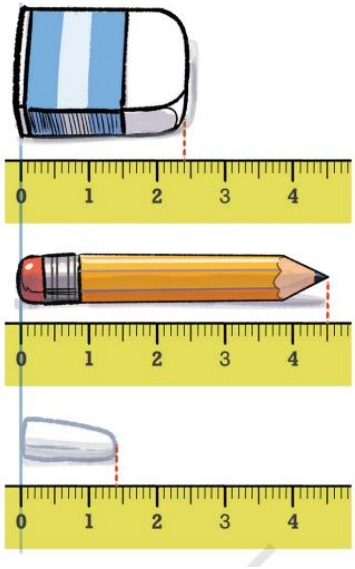
## Readable fraction and diagram answers

The supplied Mathematics chapter contains **20 supplementary answer pages** after its lesson pages. Because extracting stacked fractions into plain text removes the numerator–denominator layout, the original answer pages are included here so no fractions, measurements or number-line diagrams are misrepresented. The **key worked answers above** and **lesson/practice notes below** remain fully editable in LaTeX. Use the textbook diagram whenever a question depends on the positions of marks on a scale or number line. Some supplied companion answers may require checking.

# Chapter – 3
# A Peek Beyond the Point

**Page No. 47**

**Write the measurements of the objects shown in the picture.**




**Ans:**

$$\text{Eraser} \rightarrow 2\frac{4}{10}\ \text{cm}, \text{Pencil} \rightarrow 4\frac{5}{10}\ \text{cm} \rightarrow 4\frac{1}{2}\ \text{cm}, \text{Chalk} \rightarrow 1\frac{4}{10}\ cm$$

**Page No. 49**

**Arrange the lengths in increasing order:**

**(a)** $\frac{9}{10}$ **(b)** $1\frac{7}{10}$ **(c)** $\frac{130}{10}$ **(d)** $13\frac{1}{10}$

**(e)** $10\frac{5}{10}$ **(f)** $7\frac{6}{10}$ **(g)** $6\frac{7}{10}$ **(h)** $\frac{4}{10}$

**Ans:**

$$\frac{4}{10}, \frac{9}{10}, 1\frac{7}{10}, 6\frac{7}{10}, 7\frac{6}{10}, 10\frac{5}{10}, \frac{130}{10}, 13\frac{1}{10}$$

**Page No. 50**

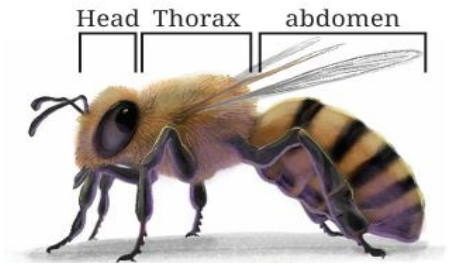
**Arrange the following lengths in increasing order:** $4\frac{1}{10}$ , $\frac{4}{10}$ , $\frac{41}{10}$ , $41\frac{1}{10}$

**Ans:**

Increasing order: $\frac{4}{10}$ , $\frac{41}{10}$ = $4\frac{1}{10}$ , $41\frac{1}{10}$

**Page No. 51**

**The lengths of the body parts of a honeybee are given. Find its total length.**

| | |
|---|---|
|   | Head $= 2\frac{3}{10}$ units<br>Thorax $= 5\frac{4}{10}$ units<br>Abdomen $= 7\frac{5}{10}$ units |

**Ans:** $15\frac{2}{10}$ units.

**Page No. 52**

**A Celestial Pearl Danio's length is $2\frac{4}{10}$ cm, and the length of a Philippine Goby is $\frac{9}{10}$ cm. What is the difference in their lengths?**

**Ans:** $1\frac{5}{10}$ cm

**Observe the given sequences of numbers. Identify the change after each term and extend the pattern:**

**(a)** **4, $4\frac{3}{10}$, $4\frac{6}{10}$,** _____, _____, _____, _____

**(b)** **$8\frac{2}{10}$, $8\frac{7}{10}$, $9\frac{2}{10}$,** _____, _____, _____, _____

**(c)** **$7\frac{6}{10}$, $8\frac{7}{10}$,** _____, _____, _____, _____

**(d)** $5\frac{7}{10}, 5\frac{3}{10},$ _____, _____, _____, _____

**(e)** $13\frac{5}{10}, 13, 12\frac{5}{10},$ _____, _____, _____, _____

**(f)** $11\frac{5}{10}, 10\frac{4}{10}, 9\frac{3}{10},$ _____, _____, _____, _____

**Ans:**

(a) $4, 4\frac{3}{10}, 4\frac{6}{10}, 4\frac{9}{10}, 5\frac{2}{10}, 5\frac{5}{10}, 5\frac{8}{10}$ → increment of $\frac{3}{10}$ each time.

(b) $8\frac{2}{10}, 8\frac{7}{10}, 9\frac{2}{10}, 9\frac{7}{10}, 10\frac{2}{10}, 10\frac{7}{10}, 11\frac{2}{10}$ → increment of $\frac{5}{10}$ each time.

(c) $7\frac{6}{10}, 8\frac{7}{10}, 9\frac{8}{10}, 10\frac{9}{10}, 12, 13\frac{1}{10}$ → increment of $1 + \frac{1}{10}$

(d) $5\frac{7}{10}, 5\frac{3}{10}, 4\frac{9}{10}, 4\frac{5}{10}, 4\frac{1}{10}, 3\frac{7}{10}$ → subtracting: $\frac{4}{10}$ each time

(e) $13\frac{5}{10}, 13, 12\frac{5}{10}, 12, 11\frac{5}{10}, 11, 10\frac{5}{10}$ → decreasing by $\frac{5}{10}$

(f) $11\frac{5}{10}, 10\frac{4}{10}, 9\frac{3}{10}, 8\frac{2}{10}, 7\frac{1}{10}, 6$ → decreasing by $1\frac{1}{10}$

**Page No. 53**

? **How many one-hundredths make one-tenth? Can we also say that the length is 4 units and 45 one-hundredths?**

**Ans:** $\frac{1}{10} = \frac{10}{100}$

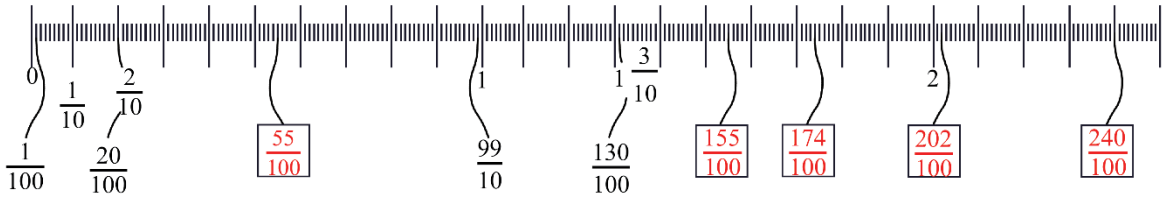
So, 10 one-hundredths make one-tenth.

Yes, we can write it $4 + \frac{45}{100}$ = 4 units and 45 one-hundredths.

**Page No. 54**

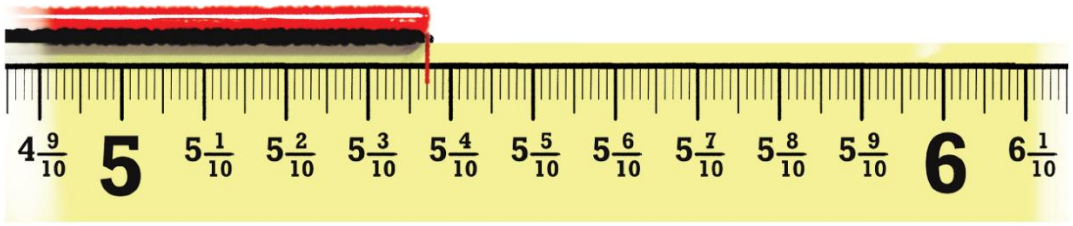
**Observe the figure below. Notice the markings and the corresponding lengths written in the boxes when measured from 0. Fill the lengths in the empty boxes.**

**Ans:**



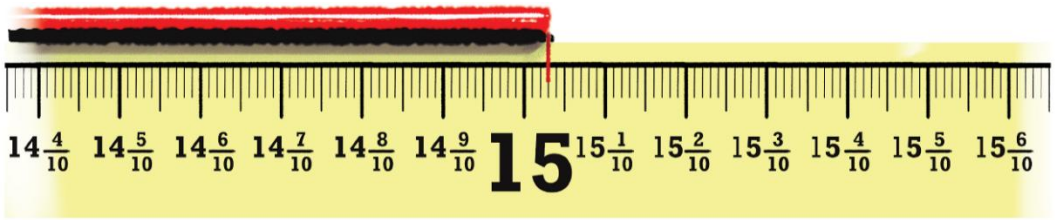

**For the lengths shown below write the measurements and read out the measures in words.**

**Ans:**




Measurement $-5\frac{37}{100}$

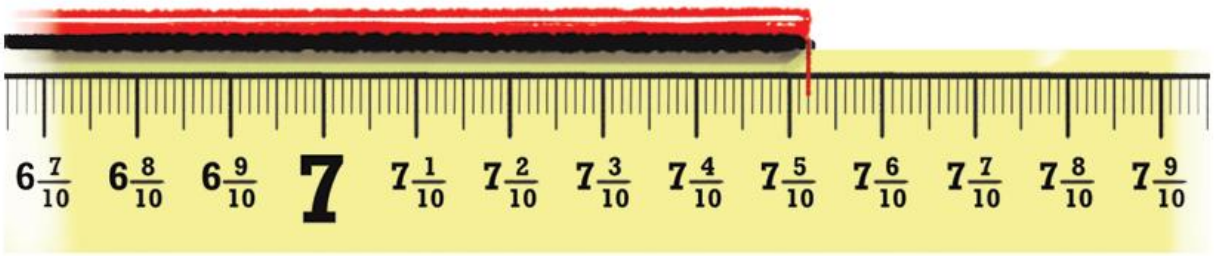
In words: Five and thirtyseven-hundredths.




Measurement $-15\frac{3}{100}$

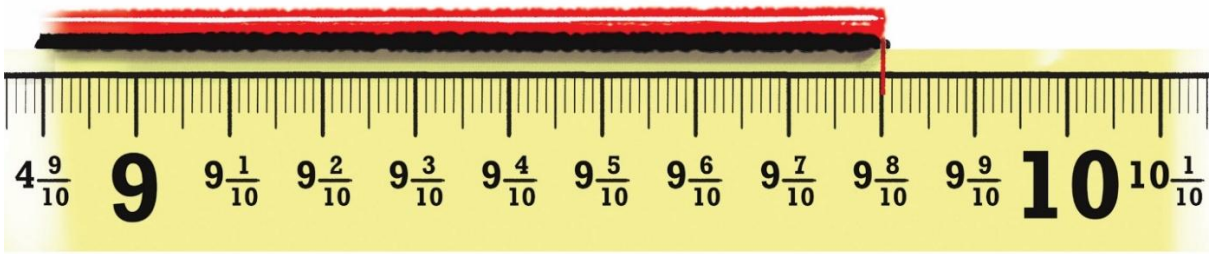
In words: Fifteen and three-hundredths.

**Page No. 55**




Measurement $-7\frac{52}{100}$

In words: Seven and fiftytwo-hundredths.



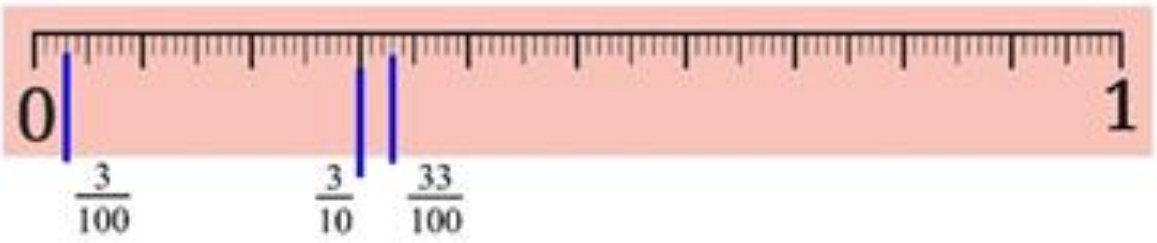

Measurement $-9\frac{80}{100}$

In words: Nine and eighty-hundredths.

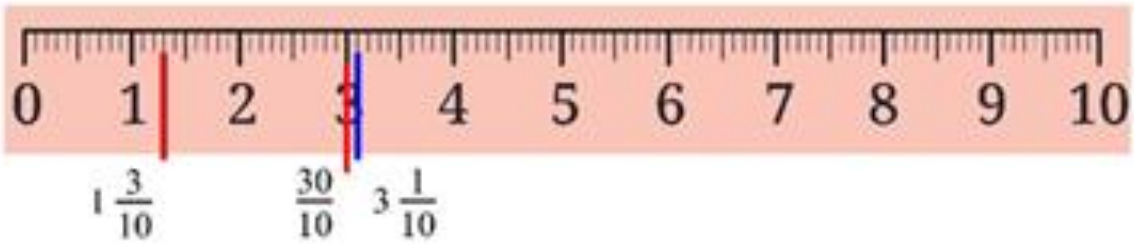
? **In each group, identify the longest and the shortest lengths. Mark each length on the scale.**

**Ans:**

(a) $\frac{3}{10}, \frac{3}{100}, \frac{33}{100}$

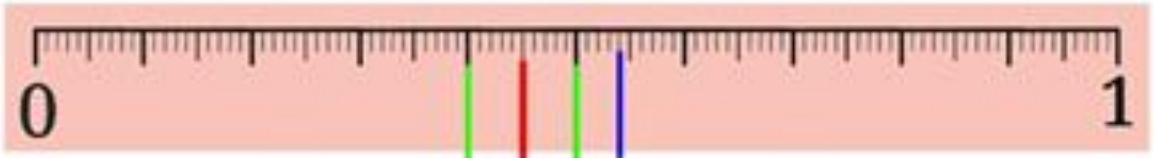


(b) $3\frac{1}{10}, \frac{30}{10}, 1\frac{3}{10}$



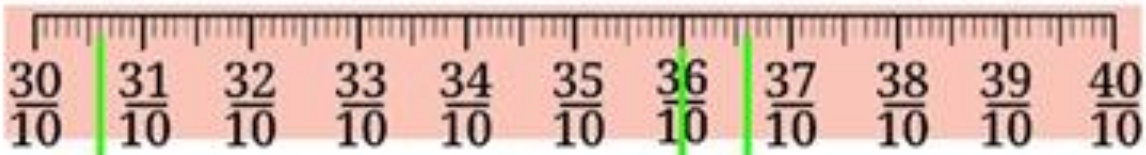

$1\frac{3}{10}$ $\frac{30}{10}$ $3\frac{1}{10}$

(c) $\frac{45}{100}, \frac{54}{100}, \frac{5}{10}, \frac{4}{10}$




$\frac{4}{10}$ $\frac{45}{100}$ $\frac{5}{10}$ $\frac{54}{100}$

(d) $3\frac{6}{10}, 3\frac{6}{100}, 3\frac{6}{10}\frac{6}{100}$




$3\frac{6}{100}$ $3\frac{6}{10}$ $3\frac{6}{10}\frac{6}{100}$

(a) Longest = $\frac{33}{100}$ Shortest = $\frac{3}{100}$

(b) Longest = $3\frac{1}{10}$ Shortest = $1\frac{3}{10}$

(c) Longest = $\frac{54}{100}$ Shortest = $\frac{4}{10}$

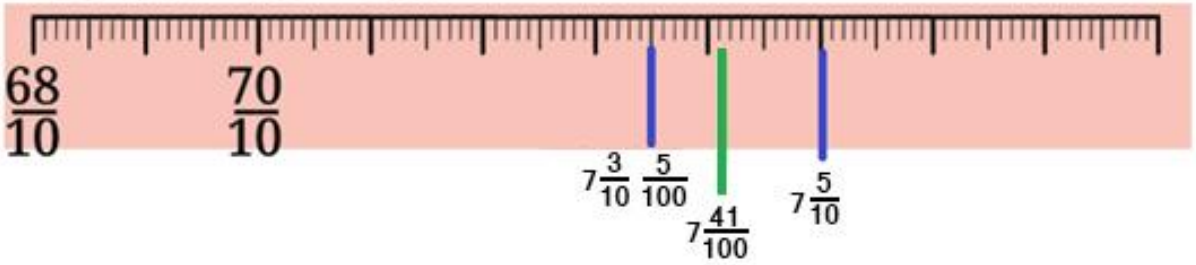
(d) Longest = $3\frac{6}{10}\ \frac{6}{100}$ Shortest = $3\frac{6}{100}$

**Page No. 56**

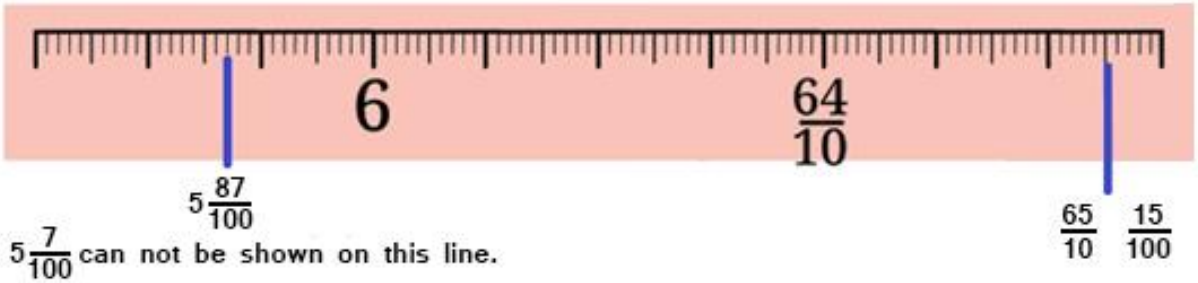
(e) $\frac{8}{10}\ \frac{2}{100}, \frac{9}{100}, 1\frac{8}{100}$




(f) $7\frac{3}{10}\ \frac{5}{100}, 7\frac{5}{10}, 7\frac{41}{100}$




(g) $\frac{65}{10}\ \frac{15}{100}, 5\frac{87}{100}, 5\frac{7}{100}$




(e) Longest = $1\frac{8}{100}$ Shortest = $\frac{9}{100}$

(f) Longest = $7\frac{5}{100}$ Shortest = $7\frac{3}{10}\ \frac{5}{100}$

(g) Longest = $\frac{65}{10}\ \frac{15}{100}$ Shortest = $5\frac{7}{100}$

**Page No. 58**

**Figure it Out**

**Find the sums and differences**

(a) $\frac{3}{10} + 3\ \frac{4}{100}$

**Ans:** $3\frac{34}{100}$

(b) $9\frac{5}{10}\ \frac{7}{100} + 2\frac{1}{10}\frac{3}{100}$

**Ans:** $11\frac{7}{10}$

(c) $15\frac{6}{10}\frac{4}{100} + 14\frac{3}{10}\frac{6}{100}$

**Ans:** 30

(d) $7\frac{7}{100} - 4\frac{4}{100}$

**Ans:** $3\frac{3}{100}$

(e) $8\frac{6}{100} - 5\frac{3}{100}$

**Ans:** $3\frac{3}{100}$

(f) $12\frac{6}{10}\ \frac{2}{100} - \frac{9}{10}\ \frac{9}{100}$

**Ans:** $11\ \frac{63}{100}$

**Page No. 63**

**Make a place value table similar to the one above. Write each quantity in decimal form and in terms of place value, and read the number:**

**Ans:**

| Question | | Hundreds | Tens | Ones | Tenths | Hundredths | Thousandth | Decimal Form | Read as |
|---|---|---|---|---|---|---|---|---|---|
| a | 2 ones, 3 tenths and 5 hundredths | – | – | 2 | 3 | 5 | – | $2 + \frac{3}{10} + \frac{5}{100}$ = 2.35 | Two point three five (or two and thirty-five hundredths) |
| b | 1 ten and 5 tenths | – | 1 | – | 5 | – | – | $10 + \frac{5}{10}$ = 10.5 | Ten point five (or ten and five tenths) |
| c | 4 ones and 6 hundredths | – | – | 4 | – | 6 | – | $4 + 0 + \frac{6}{100}$ = 4.06 | Four point zero six (or four and six hundredths) |
| d | 1 hundred, 1 one and 1 hundredth | 1 | – | 1 | – | 1 | – | $100 + 0 + 1 + 0 + \frac{1}{100}$ =101.01 | One hundred one point zero one (or one hundred one and one hundredth) |
| e | $\frac{8}{100}$ and $\frac{9}{10}$ | – | – | – | 9 | 8 | – | $\frac{9}{10} + \frac{8}{100}$ = 0.98 | Zero point nine eight (or ninety-eight hundredths) |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| f | $\frac{5}{100}$ | – | – | – | | 5 | – | $\frac{5}{100} = 0.05$ | Zero point zero five (or five hundredths) |
| g | $\frac{1}{10}$ | – | – | – | 1 | – | – | $\frac{1}{10} = 0.1$ | Zero point one (or one tenth) |
| h | $2\frac{1}{10}$, $4\frac{1}{10}$, and $7\frac{7}{1000}$ | | – | 2<br>4<br>7 | 0<br>1<br>- | 1<br>–<br>- | -<br>-<br>7 | 2.01<br>$4 + \frac{1}{10} = 4.1$<br>$7 + \frac{0}{10} + \frac{0}{100} + \frac{7}{1000} = 7.007$ | Two point zero one<br>and<br>Four point one<br>and<br>Seven point zero zero seven |
| | | | | | | | | 13.117 | Thirteen point one one seven. |

**Page No. 64**

**Write these quantities in decimal form:**

**(a) 234 hundredths**

**Ans:** 2.34

**(b) 105 tenths.**

**Ans:** 10.5

**Fill in the blanks below (mm < – > cm)**

**Ans:**

| | | |
|---|---|---|
| 12 mm= 1.2 cm | 56 mm = 5.6 cm | 70 mm = 7.0 cm |
| 9 mm = 0.9 cm | 134 mm = 13.4 cm | 2036 mm = 203.6 cm |

**Page No. 66**

**Fill in blanks (cm <-> m):**

| | | |
|---|---|---|
| 36 cm = 0.36 m | 50 cm = 0.5 m | 89 cm = 0.89 m |
| 4 cm = 0.04 m | 325 cm = 3.25 m | 207 cm = 2.07 m |

**How many mm does 1 meter have?**

**Ans:** 1m = 1000 mm

**Can we write 1 mm = $\frac{1}{1000}$ m?**

**Ans:** Yes

**Page No. 68**

**Fill in the blanks below (g < – > kg)**

**Ans:**

| | | |
|---|---|---|
| 465 g = 0.465 kg | 68 g = 0.068 kg | 1560 g = 1.56 kg |
| 704 g = 0.704 kg | 560 g = 0.56 kg | 2500 g = 2.5 kg |

**Page No. 69**

**Fill in the blanks below (rupee < – > paise)**

| | | |
|---|---|---|
| 10 p = ₹0.10 | 5 p = ₹0.05 | 36 p = ₹0.36 |
| 50 p = ₹0.50 | 99 p = ₹0.99 | 250 p = ₹2.50 |

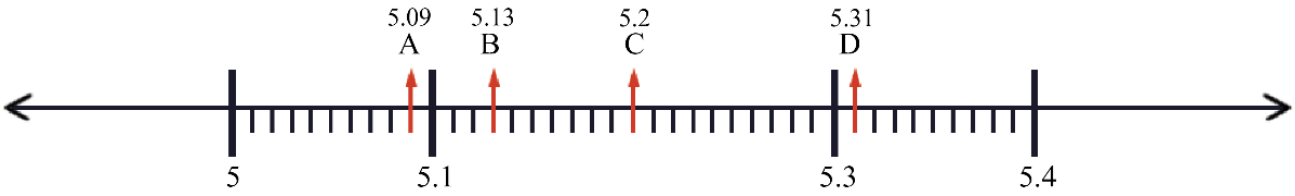
**Page No. 70**

**Name all the divisions between 1 and 1.1 on the number line.**

**Ans:** The small divisions between 1 and 1.1 on the number line (if the sub-division is of 10 parts) are: 1.01, 1.02, 1.03, 1.04, 1.05, 1.06, 1.07, 1.08, 1.09

**Identify and write the decimal numbers against the letters.**

**Ans:**




**Page No. 71**

**Can you tell which of these is the smallest and which is the largest?**

**Ans:**

| Decimal Number | Units | Tenths | Hundredths | Thousandths |
|---|---|---|---|---|
| 0.2 | 0 | 2 | 0 | 0 |
| 0.20 | 0 | 2 | 0 | 0 |
| 0.200 | 0 | 2 | 0 | 0 |
| 0.02 | 0 | 0 | 2 | 0 |
| 0.002 | 0 | 0 | 0 | 2 |

0.2 = 0.20 = 0.200 is the largest.

0.002 is the smallest.

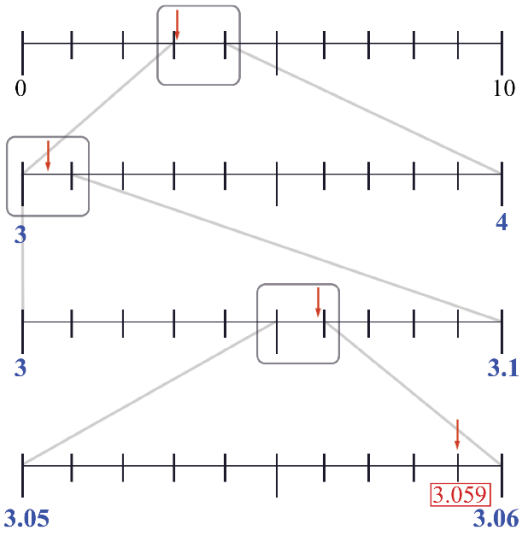
**Which of these are the same: 4.5, 4.05, 0.405, 4.050, 4.50, 4.005, 04.50?**

**Ans:** 4.5, 4.50, and 04.50 are equal.

4.05 and 4.050 are equal.

**Identify the decimal number in the last number line in Figure (b) denoted by '?'**

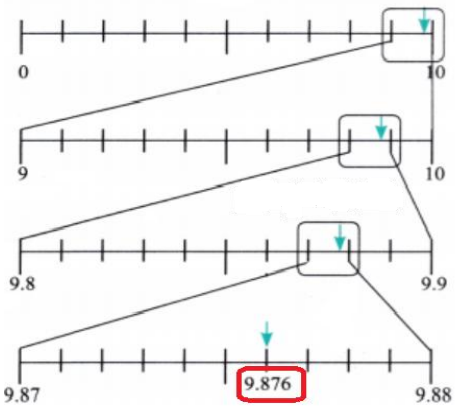
**Ans:**

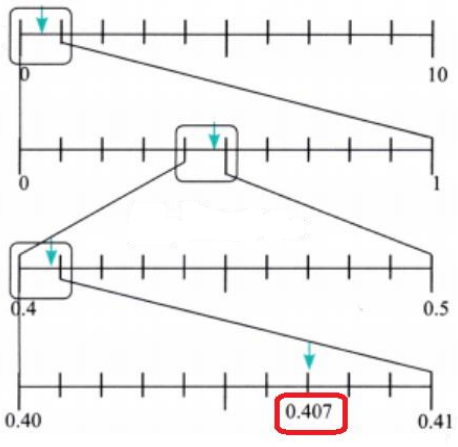



(b)

**Make such number lines for the decimal numbers: (a) 9.876 (b) 0.407.**

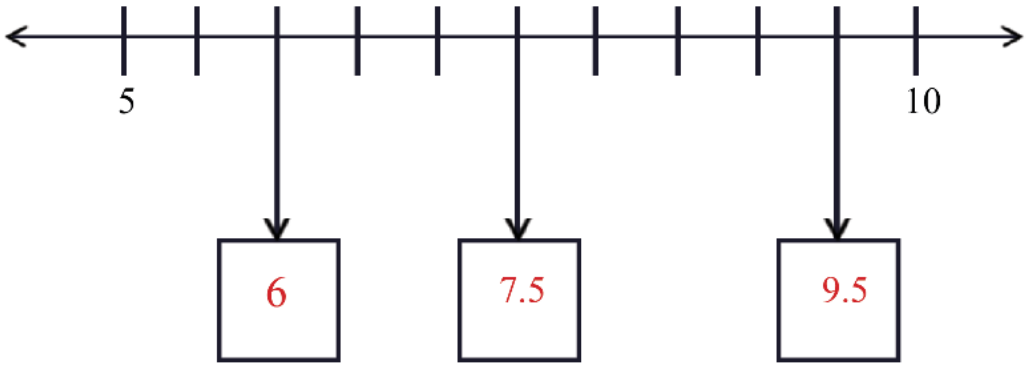
**Ans:** **(a)** **(b)**



**In the number line shown below, what decimal numbers do the boxes labelled ‘a’, ‘b’, and ‘c’ denote?**

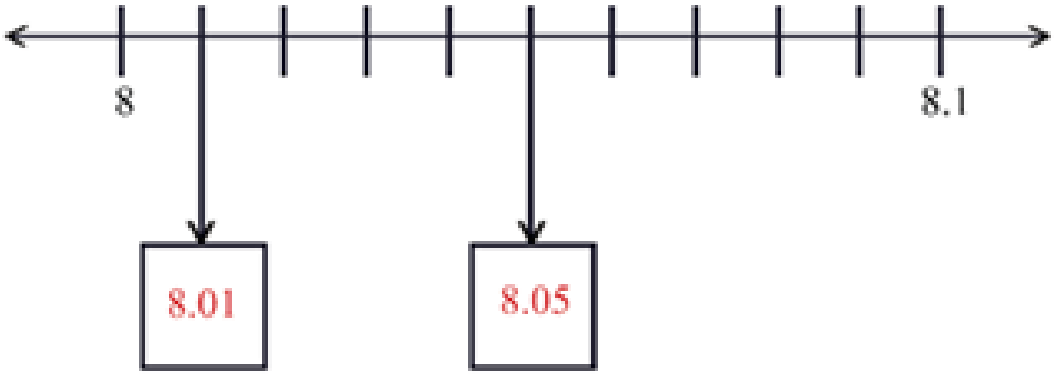
**Ans:**

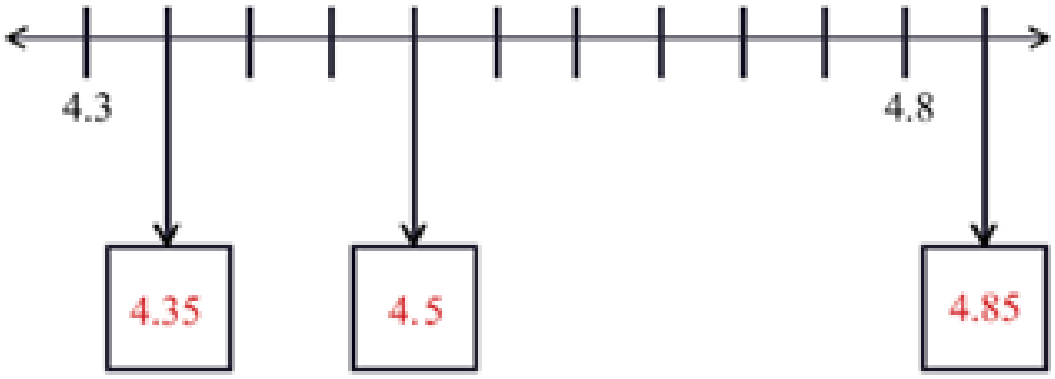



**Page No. 72**

**Using similar reasoning find out the decimal numbers in the boxes below.**

**Ans:**



**Page No. 73**

**Which decimal number is greater?**
**(a) 1.23 or 1.32 (b) 3.81 or 13.800 (c) 1.009 or 1.090**

**Ans:**

(a) 1.32 is larger.

(b) 13.800 is larger.

(c) 1.090 is larger.

**Consider the decimal numbers 0.9, 1.1, 1.01 and 1.11**
**Which of the above is closest to 1.09?**

**Ans:** 1.1 is closest to 1.09.

**Which among these is closest to 4: 3.56, 3.65, 3.099?**

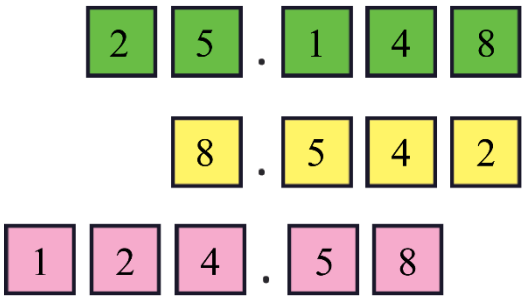
**Ans:** 3.65 is closest to 4.

**Which among these is closest to 1: 0.8, 0.69, 1.08?**

**Ans:** 1.08 is closest to 1.

**In each case below use the digits 4, 1, 8, 2, and 5 exactly once and try to make a decimal number as close as possible to 25.**

**Ans:**




**Page No. 75**

**Figure it Out**

**1. Find the sums**

**(a) 5.3 + 2.6**
**(b) 18 + 8.8**
**(c) 2.15 + 5.26**
**(d) 9.01 + 9.10**
**(e) 29.19 + 9.91**
**(f) 0.934 + 0.6**
**(g) 0.75 + 0.03**
**(h) 6.236 + 0.487**

**Ans:**

(a) 7.9 (b) 26.8
(c) 7.41 (d) 18.11
(e) 39.10 (f) 1.534
(g) 0.78 (h) 6.723

**2. Find the differences**

**(a) 5.6 – 2.3** **(b) 18 – 8.8**
**(c) 10.4 – 4.5** **(d) 17 – 16.198**
**(e) 17 – 0.05** **(f) 34.505 – 18.1**
**(g) 9.9 – 9.09** **(h) 6.236 – 0.487**

**Ans:**

(a) 3.3 (b) 9.2
(c) 5.9 (d) 0.802
(e) 16.95 (f) 16.405
(g) 0.81 (h) 5.749

**Continue this sequence 4.4, 4.8. 5.2, 5.6, 6.0, … and write the next 3 terms.**

**Ans:** 4.4, 4.8. 5.2, 5.6, 6.0, 6.4, 6.8, 7.2

**Page No. 76**

**Similarly, identify the change and write the next 3 terms for each sequence given below.**
**Try to do this computation mentally.**

**(a) 4.4, 4.45, 4.5, …** **(b) 25.75, 26.25, 26.75, …**
**(c) 10.56, 10.67, 10.78, …** **(d) 13.5, 16, 18.5, …**
**(e) 8.5, 9.4, 10.3, …** **(f) 5, 4.95, 4.90, …**
**(g) 12.45, 11.95, 11.45, …** **(h) 36.5, 33, 29.5, …**

**Ans:**

(a) 4.4, 4.45, 4.5,4.55, 4.6, 4.65  increment of + 0.05

(b) 25.75, 26.25, 26.75, 27.25, 27.75, 28.25  add 0.5

(c) 10.56, 10.67, 10.78, 10.89, 11.0, 11.11 Add 0.11

(d) 13.5, 16, 18.5, 21.0, 23.5, 26.0  Add 2.5

(e) 8.5, 9.4, 10.3, 11.2, 12.1, 13.0 → Add 0.9

(f) 5, 4.95, 4.90, 4.85, 4.80, 4.75 → Subtract 0.05

(g) 12.45, 11.95, 11.45, 10.95, 10.45, 9.95 → Subtract 0.50

(h) 36.5, 33, 29.5, 26.0, 22.5, 19.0 → Subtract 3.5

**Page No. 78**

**Figure it out**

**1. Convert the following fractions into decimals:**

**(a) $\frac{5}{100}$** **(b) $\frac{16}{1000}$** **(c) $\frac{12}{10}$** **(d) $\frac{254}{1000}$**

**Ans:**

(a) $\frac{5}{100} = 0.05$ (b) $\frac{16}{1000} = 0.016$ (c) $\frac{12}{10} = 1.2$ (d) $\frac{254}{1000} = 0.254$

**Page No. 79**

**2. Convert the following decimals into a sum of tenths, hundredths and thousandths:**

**(a) 0.34** **(b) 1.02** **(c) 0.8** **(d) 0.362**

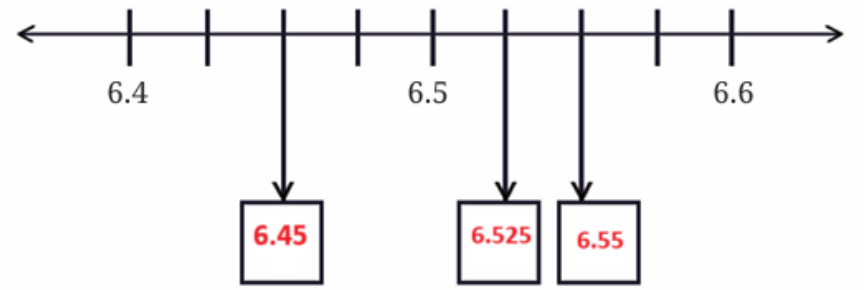
**Ans:**

(a) $\frac{3}{10} + \frac{4}{100}$ (b) $\frac{100}{100} + \frac{2}{100} = 1 + \frac{2}{100}$

(c) $\frac{8}{10}$ (d) $\frac{3}{10} + \frac{6}{100} + \frac{2}{1000}$

**3. What decimal number does each letter represent in the number line below?**

**Ans:**




**4. Arrange the following quantities in descending order:**

**(a) 11.01, 1.011, 1.101, 11.10, 1.01**

 Descending order: 11.10, 11.01, 1.101, 1.011, 1.01.

**(b) 2.567, 2.675, 2.768, 2.499, 2.698**

 Descending order: 2.768, 2.698, 2.675, 2.567, 2.499.

**(c) 4.678 g, 4.595 g, 4.600 g, 4.656 g, 4.666 g**

Descending order: 4.678, 4.666, 4.656, 4.600, 4.595.

**(d) 33.13 m, 33.31 m, 33.133 m, 33.331 m, 33.313 m**

 Descending order: 33.331, 33.313, 33.31, 33.133, 33.13.

**5. Using the digits 1, 4, 0, 8, and 6 make: (a) the decimal number closest to 30 (b) the smallest possible decimal number between 100 and 1000.**

**Ans:** Using the digits 1, 4, 0, 8, and 6, we can make:

(a) 40.168

(b) 104.68

**6. Will a decimal number with more digits be greater than a decimal number with fewer digits?**

**Ans:** No. It is not necessary. For example, 2.05 (2 digits after decimal) vs 2.5 (1 digit after decimal). 2.5 >2.05

**7. Mahi purchases 0.25 kg of beans, 0.3 kg of carrots, 0.5 kg of potatoes, 0.2 kg of capsicums, and 0.05 kg of ginger. Calculate the total weight of the items she bought.**

**Ans:** 1.3 kg

**8. Pinto supplies 3.79 L, 4.2 L, and 4.25 L of milk to a milk dairy in the first three days. In 6 days, he supplies 25 liters of milk. Find the total quantity of milk supplied to the dairy in the last three days.**

**Ans:** 12.76 L

**9. Tinku weighed 35.75 kg in January and 34.50 kg in February. Has he gained or lost weight? How much is the change?**

**Ans:** He has lost weight. The change in the weight = 1.25 kg

**10. Extend the pattern: 5.5, 6.4, 6.39, 7.29, 7.28, 8.18, 8.17, ____, _____**

**Ans:** 9.07, 9.06

**11. How many millimeters make 1 kilometer?**

**Ans:** 1000000 mm

**12. Indian Railways offers optional travel insurance for passengers who book e-tickets. It costs 45 paise per passenger. If 1 lakh people opt for insurance in a day, what is the total insurance fee paid?**

**Ans:** The total insurance fee paid for 1 lakh passengers is ₹45,000.

**13. Which is greater?**

**(a)** $\frac{10}{1000}$ **or** $\frac{1}{10}$**?**

**(b) One-hundredth or 90 thousandths?**

**(c) One-thousandth or 90 hundredths?**

**Ans:**

(a) $\frac{1}{10} > \frac{1}{100}$

(b) 90 thousandths > One-hundredth

(c) 90 hundredths > One-thousandth

**Page No. 80**

**14. Write the decimal forms of the quantities mentioned (an example is given):**

**(a) 87 ones, 5 tenths and 60 hundredths**

**(b) 12 tens and 12 tenths**

**(c) 10 tens, 10 ones, 10 tenths, and 10 hundredths**

**(d) 25 tens, 25 ones, 25 tenths, and 25 hundredths**

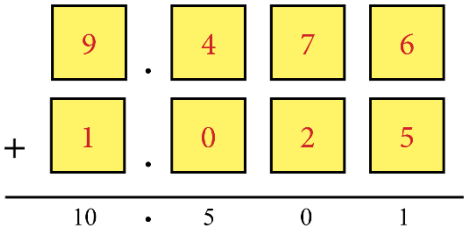
**Ans:**

(a) 88.10 121.2

(b) 111.1 277.75

**15. Using each digit 0 – 9 not more than once, fill the boxes below so that the sum is closest to 10.5:**

**Ans:**




**16. Write the following fractions in decimal form:**

**(a)** $\frac{1}{2}$ **(b)** $\frac{3}{2}$ **(c)** $\frac{1}{4}$

**(d)** $\frac{3}{4}$ **(e)** $\frac{1}{5}$ **(f)** $\frac{4}{5}$

**Ans:**

(a) 0.5 (b) 1.5 (c) 0.25

(d) 0.75 (e) 0.2 (f) 0.8

# Illustrated notes: decimal place value

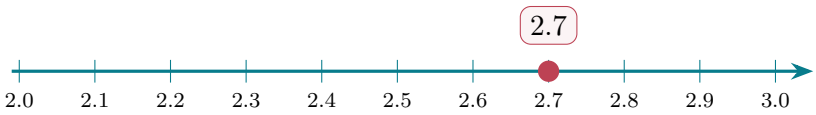
## What a decimal really means

**Key Idea**

$4.275 = 4 + \frac{2}{10} + \frac{7}{100} + \frac{5}{1000}$. The decimal point separates whole units from parts of units.

| Ones | Tenths | Hundredths | Thousandths |
|---|---|---|---|
| 4 | 2 | 7 | 5 |

## An editable number line




Each step here is one-tenth of a unit. The dot marks 2.7. We can zoom into the interval from 2.7 to 2.8 to show hundredths.

## Compare decimals by aligning places

**Worked Example**

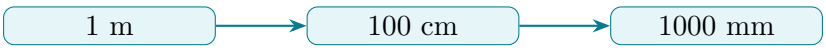
$3.508 > 3.49$ because we can write $3.49 = 3.490$ and compare tenths ($5 > 4$). Zeros added to the right of the final digit do not change a value.

## Addition and subtraction

$$\begin{array}{r} 12.45 \\ +\ 3.70 \\ \hline 16.15 \end{array} \qquad \begin{array}{r} 34.505 \\ -18.100 \\ \hline 16.405 \end{array}$$

**Remember:** Line up the **decimal points**, not the final digits. Add zeroes as placeholders.

## Measurement and money




1 kg = 1000 g; Rs. 1 = 100 paise. To convert cm to mm, multiply by 10. To convert paise to rupees, divide by 100.

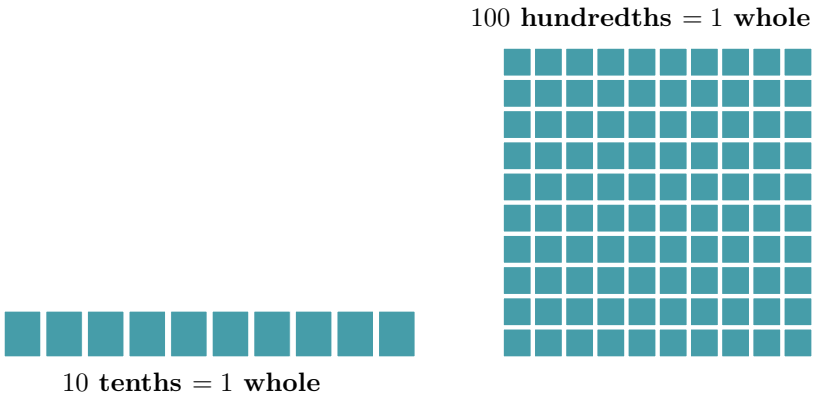
**Common Mistake**

Not every dot in a number is a decimal point: in cricket, "5.5 overs" is five overs and five balls. It is not $5\frac{5}{10}$ overs.

# Last-minute revision

- Decimal places: tenths, hundredths, thousandths.
- $0.3 = 0.30 = 0.300$, but $0.3 \neq 0.03$.
- For calculations, align decimal points vertically.
- Read units: cm, mm, m, rupees, paise and kilograms correctly.

## Mini-lab: grids for tenths and hundredths




**Zoom into the parts**

Ten strips, each one tenth of a unit, make one whole. Each strip can be divided into ten squares, making 100 hundredths in the whole. Hence $\frac{1}{10} = \frac{10}{100}$ and $0.1 = 0.10$.

## Why trailing zeroes are useful

**Worked Example**

$6.4 + 0.75 = 6.40 + 0.75 = 7.15$. Adding a zero at the **end** of 6.4 does not change its value. Writing 6.04 instead would change the value and lead to an error.

## Quick self-check (with answers)

1. $\frac{7}{100} = ?$ **0.07**.
2. 12.300 equals 12.3? **Yes**.
3. Which is larger, 0.72 or 0.7? **0.72**.
4. $8.65 - 3.8 = ?$ **4.85**.
5. 3.6 cm in mm? **36 mm**.
6. 90 paise in rupees? **Rs. 0.90**.
7. 2.125 has which decimal digits? **1 tenth, 2 hundredths, 5 thousandths**.
8. Is 0.6 or 0.59 bigger? 0.6, since $0.60 > 0.59$.

**Memory Trick**

When unsure, write the **missing trailing zeroes**: $0.9 = 0.90 = 0.900$. Then compare from left to right.